Course Code: CS1812 **Name of the Programme: B.Tech (Hons.) CSE** **Name of the Course: Discrete Mathematics and Set Theory** **Semester: 1**		
Course credit	**No. of hours per week (L + T + P)**	**Total no. of Teaching hours**
3	2+1+0	30L+15T
Course Objectives:		
a) Introduce the fundamental concepts of discrete mathematics essential for computer science and engineering.		
b) Focus on set theory principles to build a strong mathematical foundation for problem-solving in computing disciplines.		
c) Describe and analyze fundamental concepts of logic and set theory, including propositions, predicates, set operations, and relations.		
d) Explain the concepts of functions and combinatorics, emphasizing their role in the development of algorithms, data structures, and computational models.		

Course outline (Syllabus of the course)

Syllabus:

Module – 1: Sets	**No. of hours**
	4
Sets and operations on sets, Set-builder notations, types of sets, union, intersection, complement, difference, symmetric difference, their properties and laws, Venn diagrams, Cartesian Product, cardinality; multisets.	

Module – 2: Propositional Logic	**No. of hours**
	10
Propositions and truth tables, tautologies and contradictions, logical equivalence, algebra of propositions, logical implications, predicate logic.	

Module - 3: Relations and Functions	No. of hours
	6
Relations, functions-one-to-one, onto functions. The Pigeon-hole principle, function Composition; invertibility and Inverse of function.	

Module - 4: Counting Techniques and Permutations	No. of hours
	5
The Rules of Sum and Product, Permutations	

Module - 5: Combinations, Binomial Theorem s Catalan Numbers	No. of hours
	5
Combinations - The Binomial Theorem, Combinations with Repetition, The Catalan Numbers.	

Course outcomes
a) Apply the concepts of set theory to solve problems in relevant domains.
b) Apply truth tables, logical equivalences, and predicate logic to solve reasoning-based problems.
c) Apply relations, functions, and the pigeonhole principle to solve computational problems.
d) Apply principles of combinatorics to solve basic counting problems.

Text Books	
1	Rosen K, Discrete Mathematics and Its Applications, McGrawHill, 8th edition, 2021, ISBN: 978-9390727353
2	N Ch S N Iyengar, V M Chandrasekaran, Discrete Mathematics, Vikas Publishers, First Edition, 2003, ISBN :978-8125913627

Reference books	
1	Michael Huth Camp; Mark Ryan, “Logic in Computer Science: Modelling and Reasoning about Systems”, Cambridge University Press, Second 2004, ISBN-(13):978-0521543101
2	Basavaraj S Anami and Venakanna S Madalli “Discrete Mathematics – A Concept based approach”, Universities Press, 2016, ISBN-(13): 978-8173719998
3	Lipschitz, Lipson, Discrete Mathematics, McGrawHill, Second 1992, ISBN : 978-0070380318
4	G. Shanker Rao, Discrete Mathematical Structures, New Age International, 2007 ; ISBN, 8122426697, 9788122426694

Tutorials	
1	CO1: 2 Sessions
2	CO2: 5 Sessions including Activity: Introduction to Octave
3	CO3: 3 Sessions involving Activity: Implementation of Relations and Functions
4	CO4: 3 Sessions